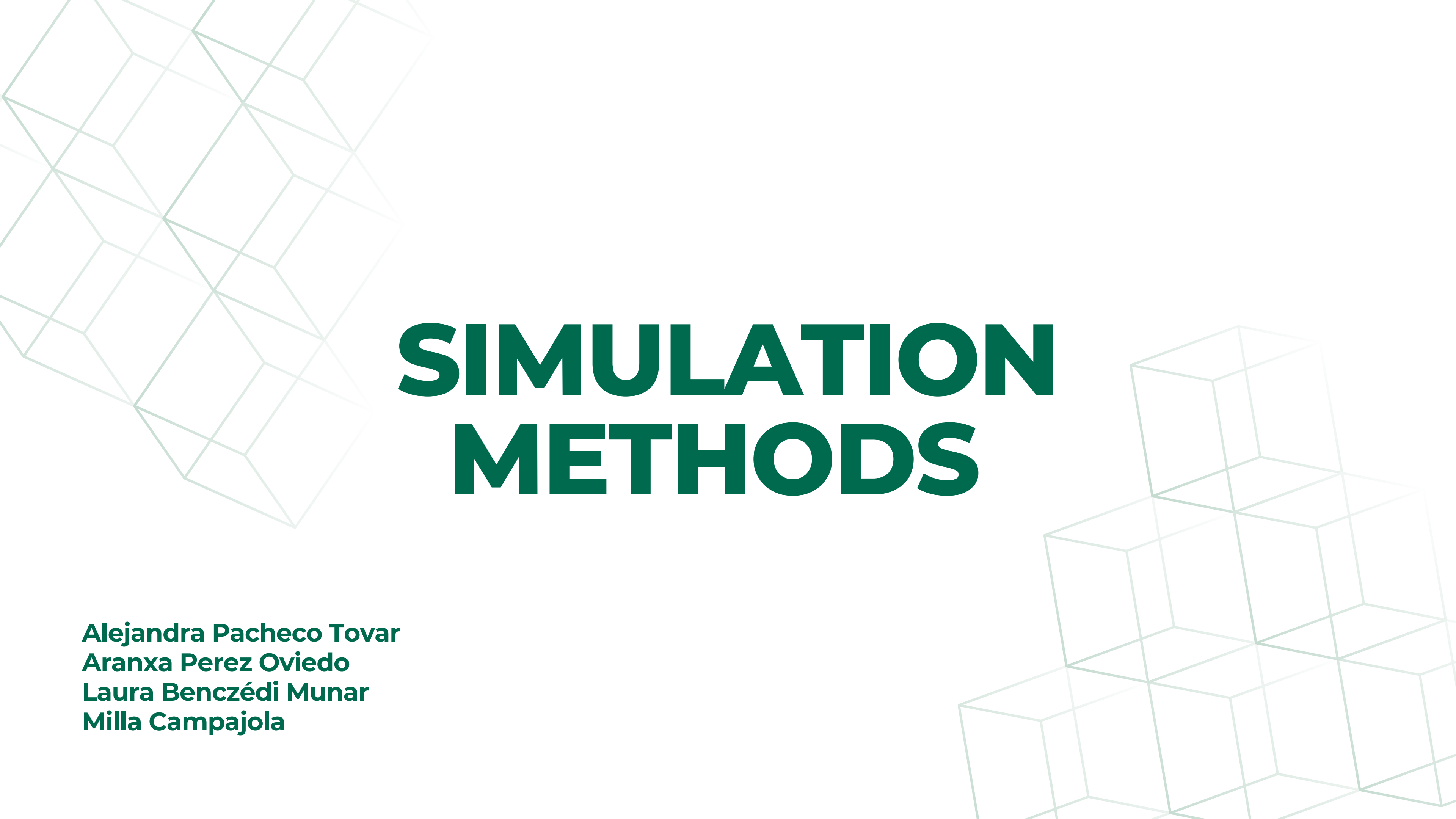




SIMULATION METHODS

**Alejandra Pacheco Tovar
Aranxa Perez Oviedo
Laura Benczédi Munar
Milla Campajola**

INTRODUCTION

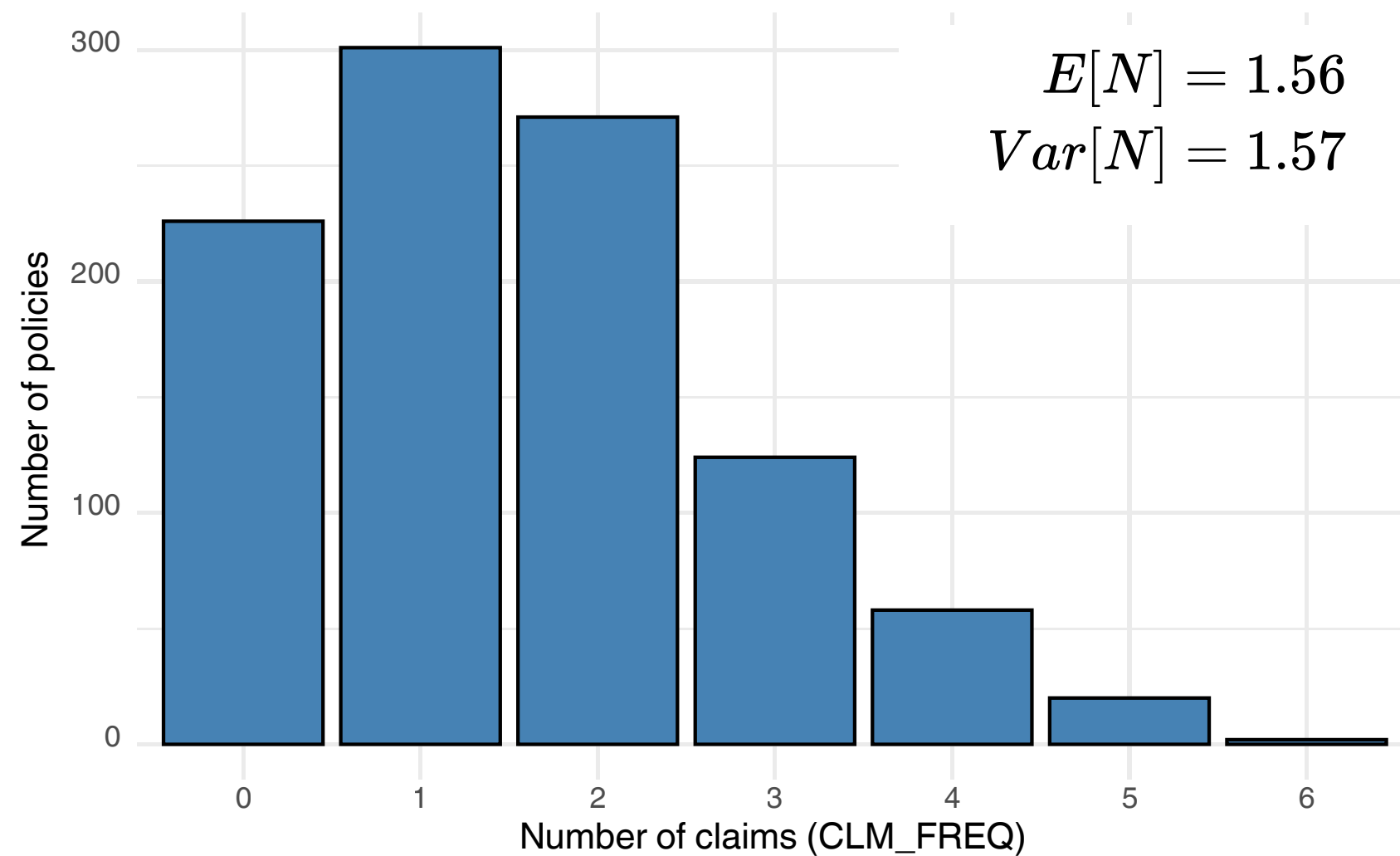
Motor insurance pricing must balance market competitiveness with portfolio profitability. **This project evaluates whether the current motor tariff can be reduced while keeping the portfolio financially sustainable**, using a classical frequency-severity framework.



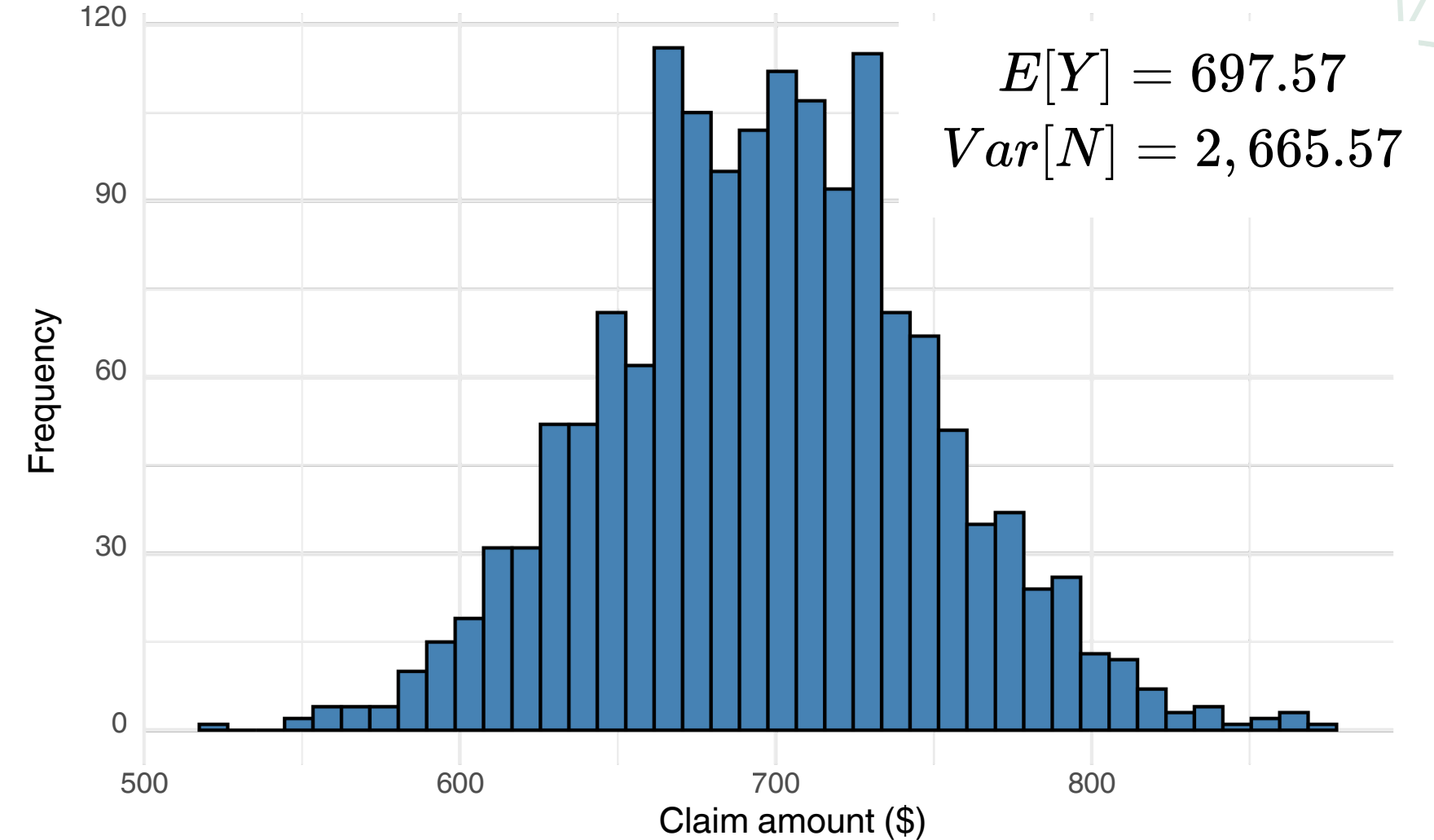
By how much, if at all, can the current tariff be safely reduced? [→](#)

EXPLORATORY DATA ANALYSIS

Distribution of claim counts (after cleaning)

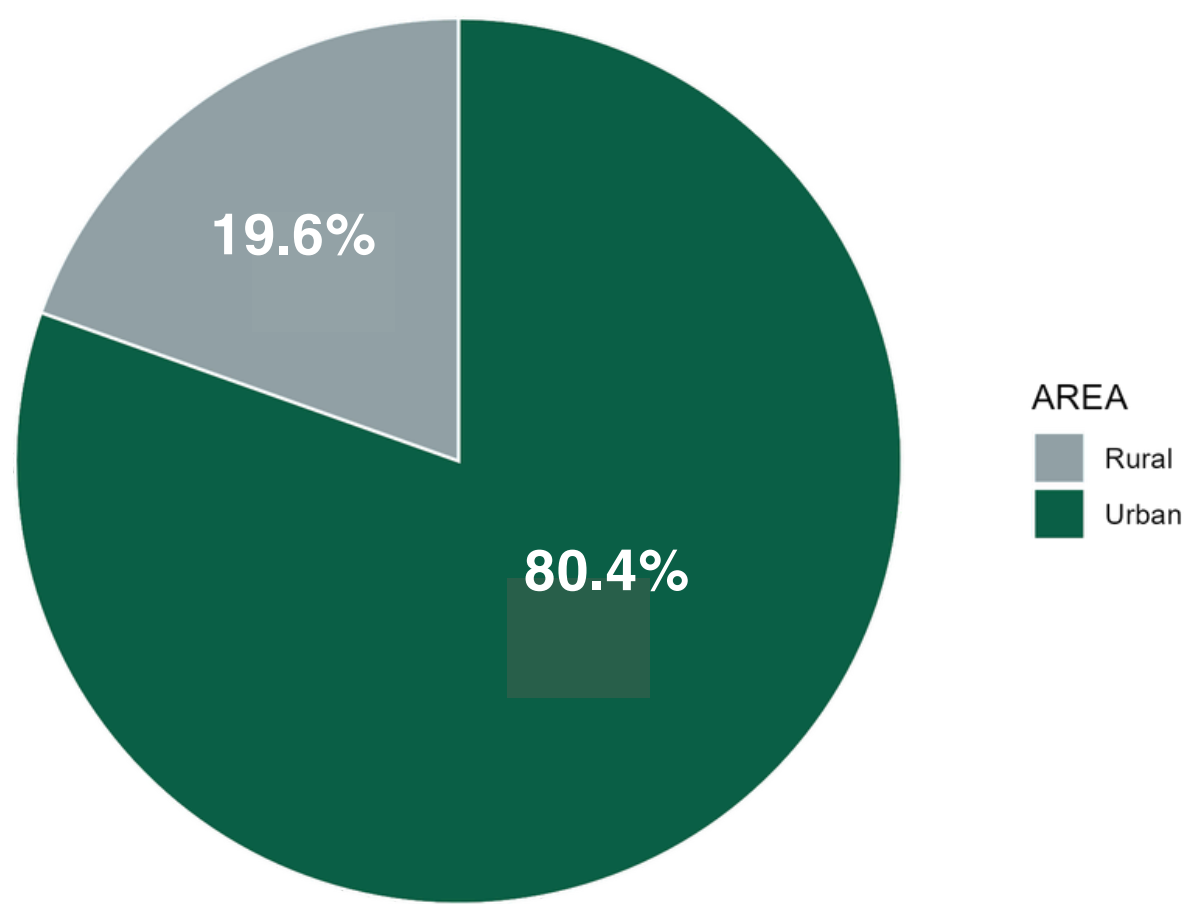


Histogram of claim severity

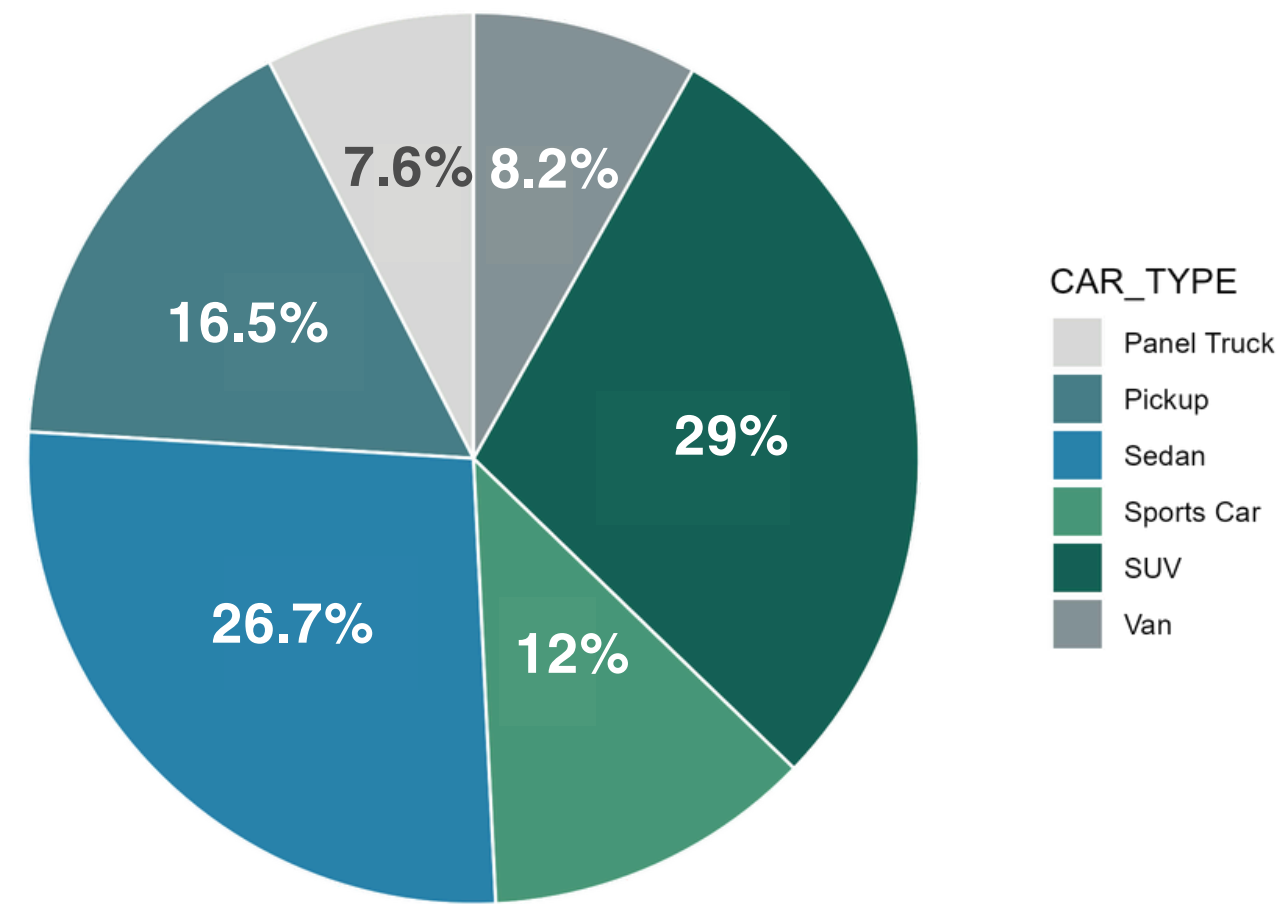


EXPLORATORY DATA ANALYSIS

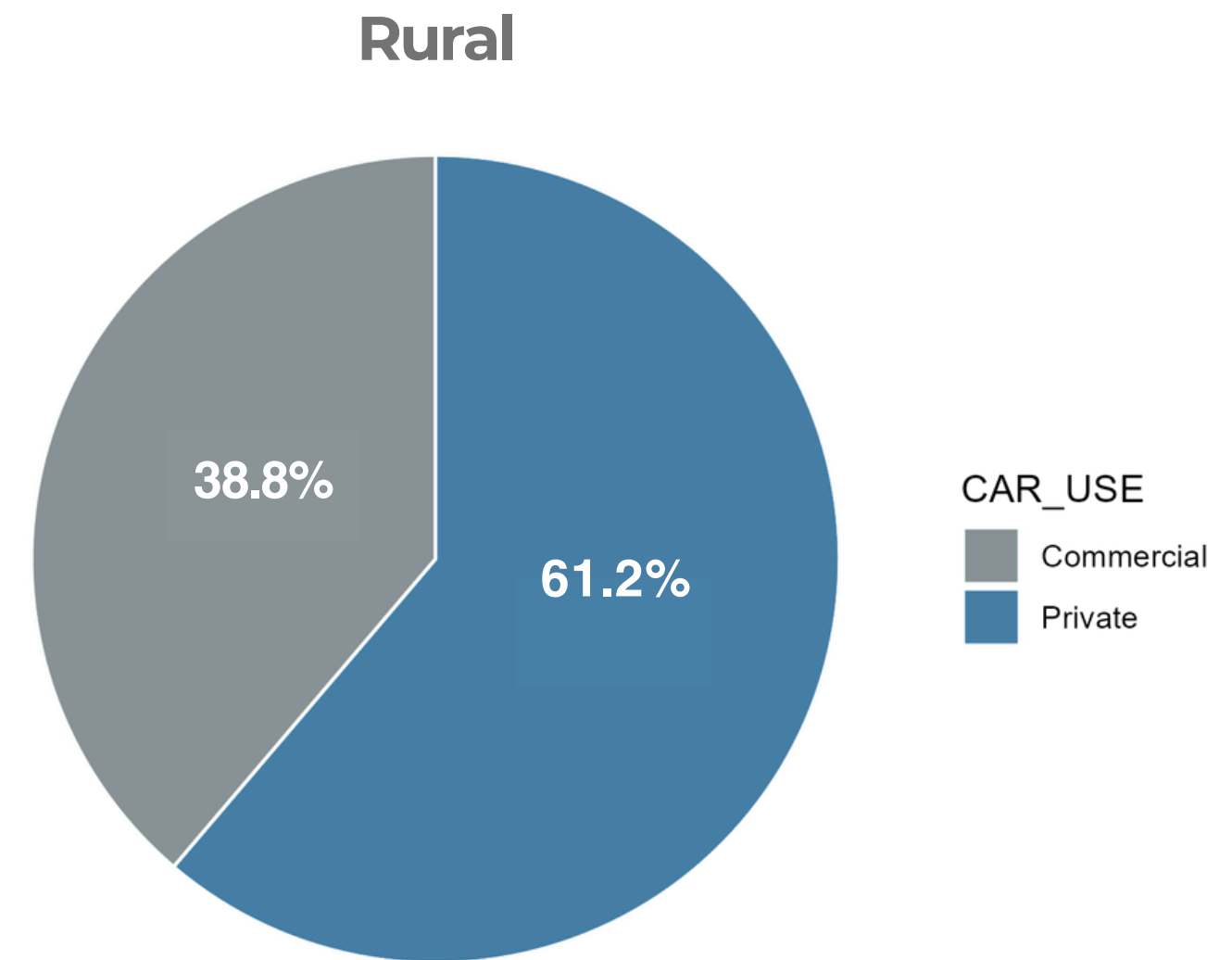
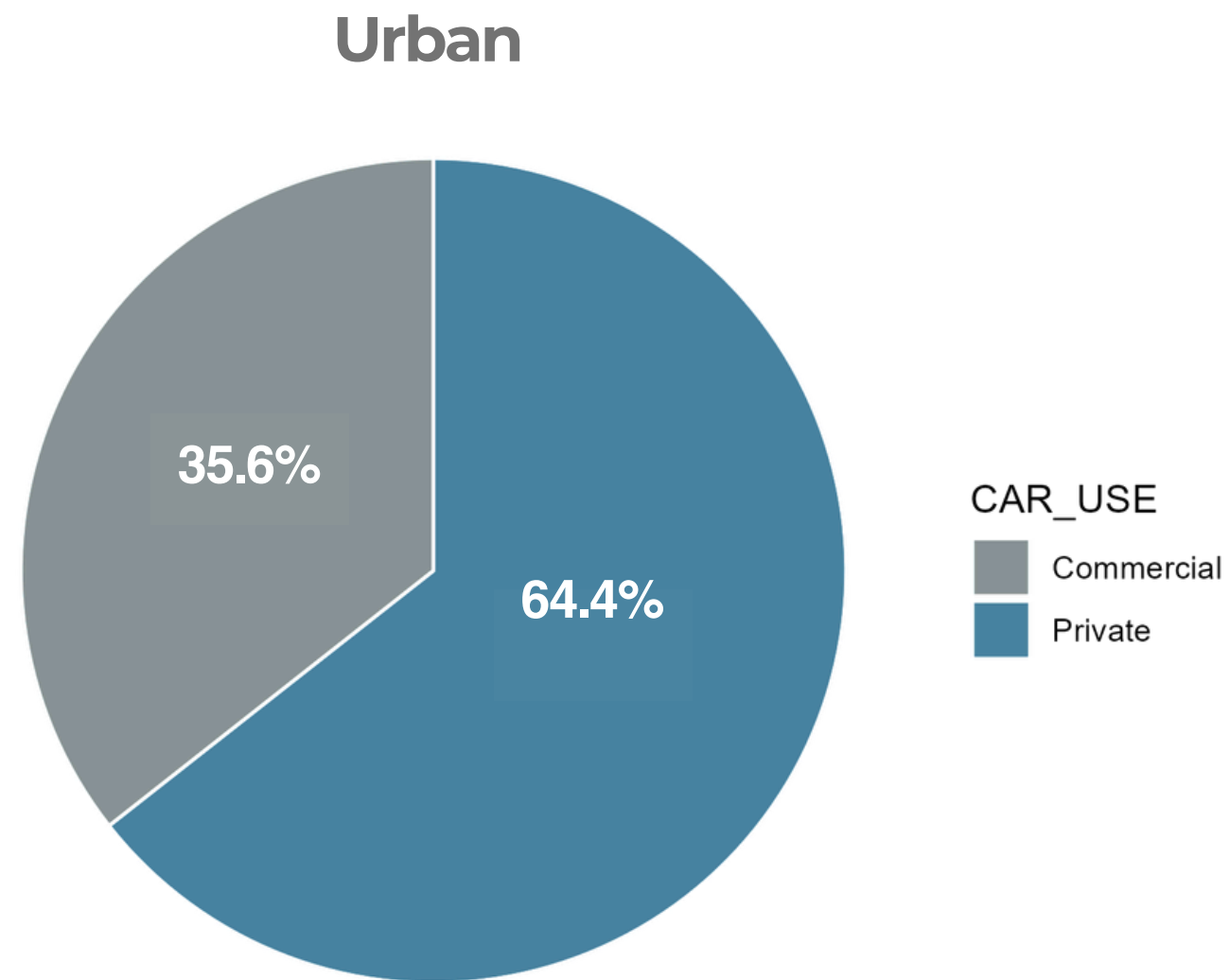
Urban vs Rural distribution



Distribution of car types



EXPLORATORY DATA ANALYSIS

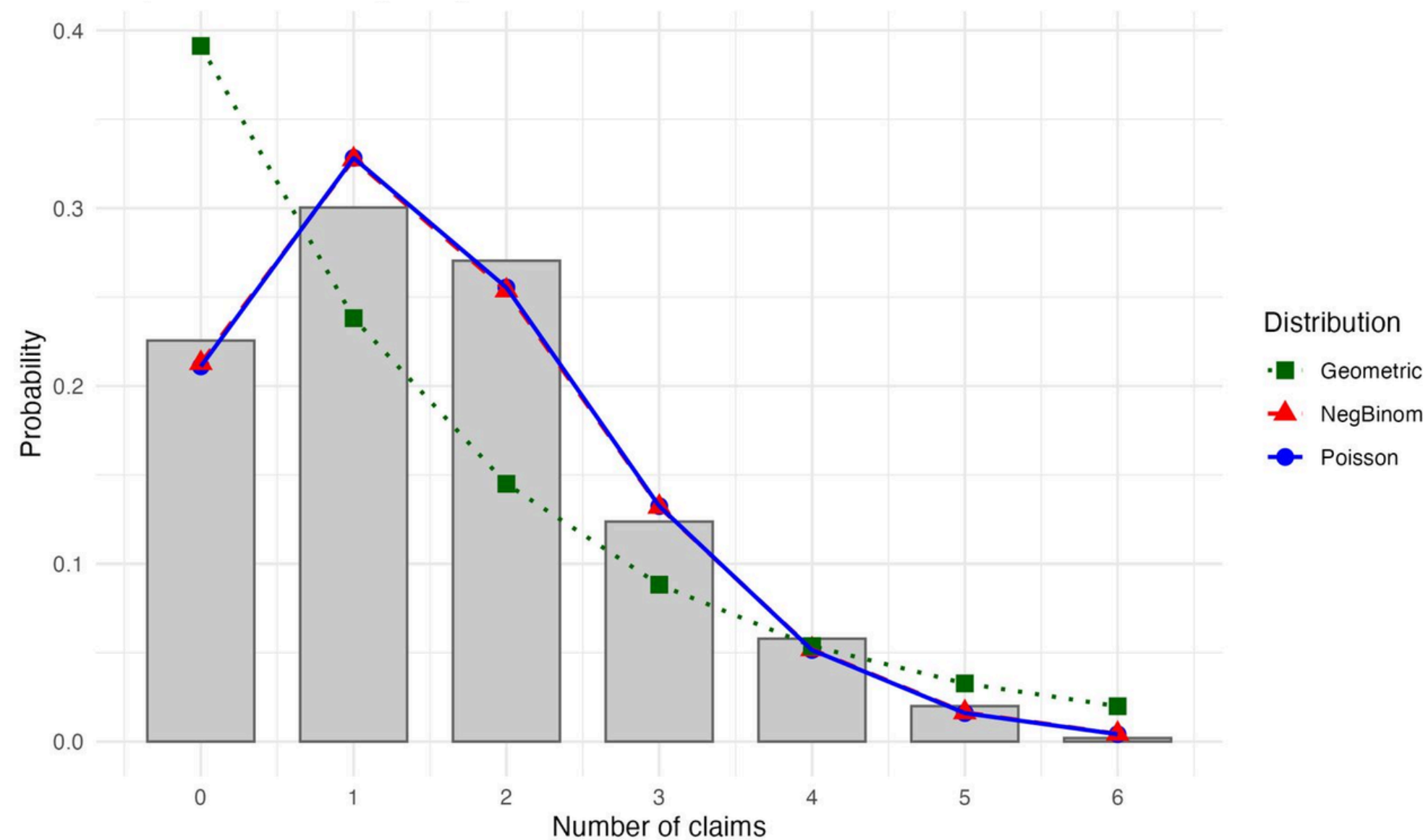


MODEL SELECTION & VALIDATION

Claim Frequency

Distribution	Parameters (MLE)	AIC
Poisson	$\lambda = 1.5559$	3148.21
Negative Binomial	size = 135.67, $\mu = 1.5559$	3150.15
Geometric	$p = 0.3913$	3430.18

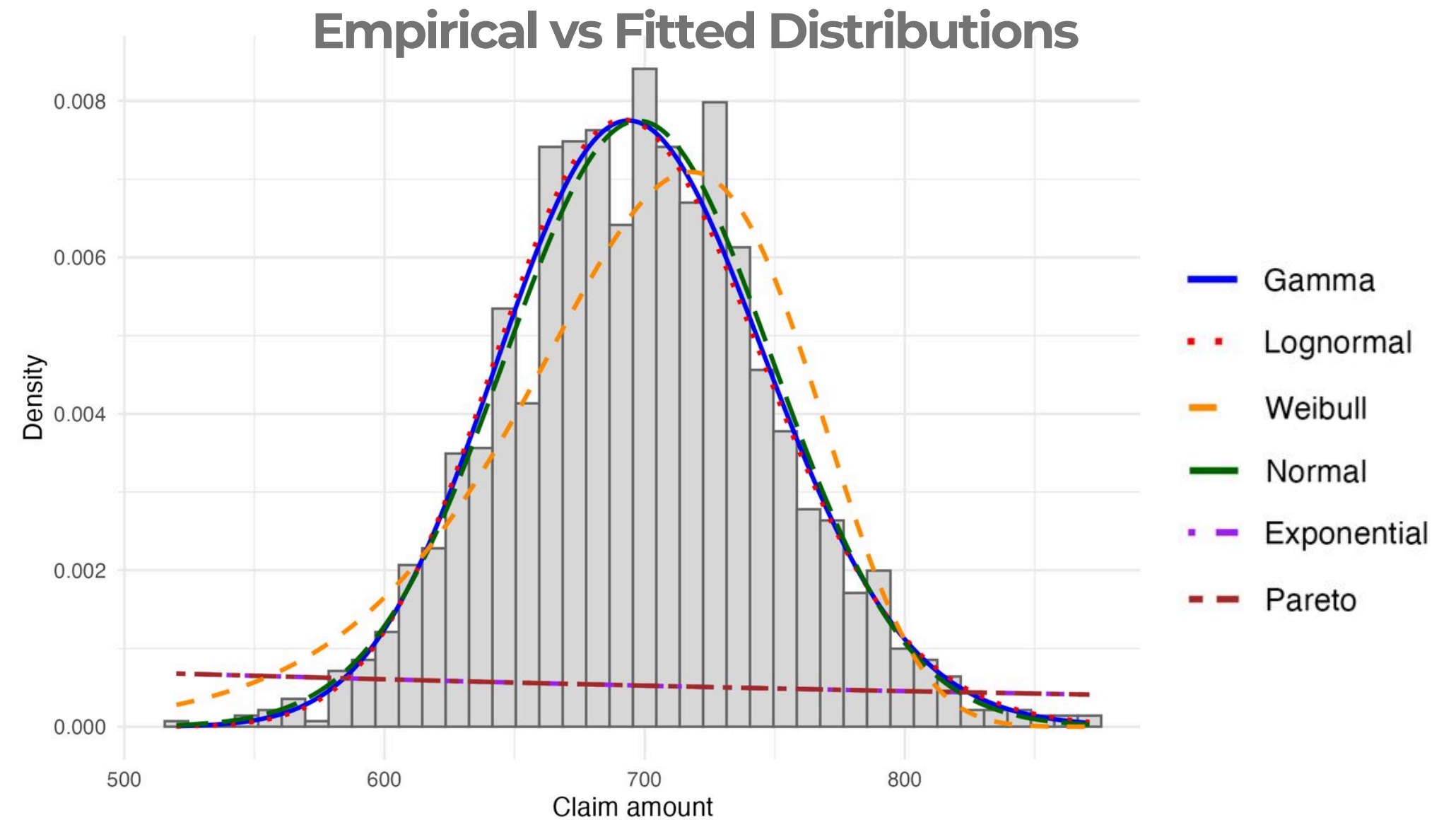
Empirical vs Fitted Distributions



MODEL SELECTION & VALIDATION

Claim Severity

Distribution	AIC
Gamma	16717.17
Lognormal	16721.05
Weibull	16913.45
Normal	16718.24
Exponential	23535.42
Pareto	23537.42

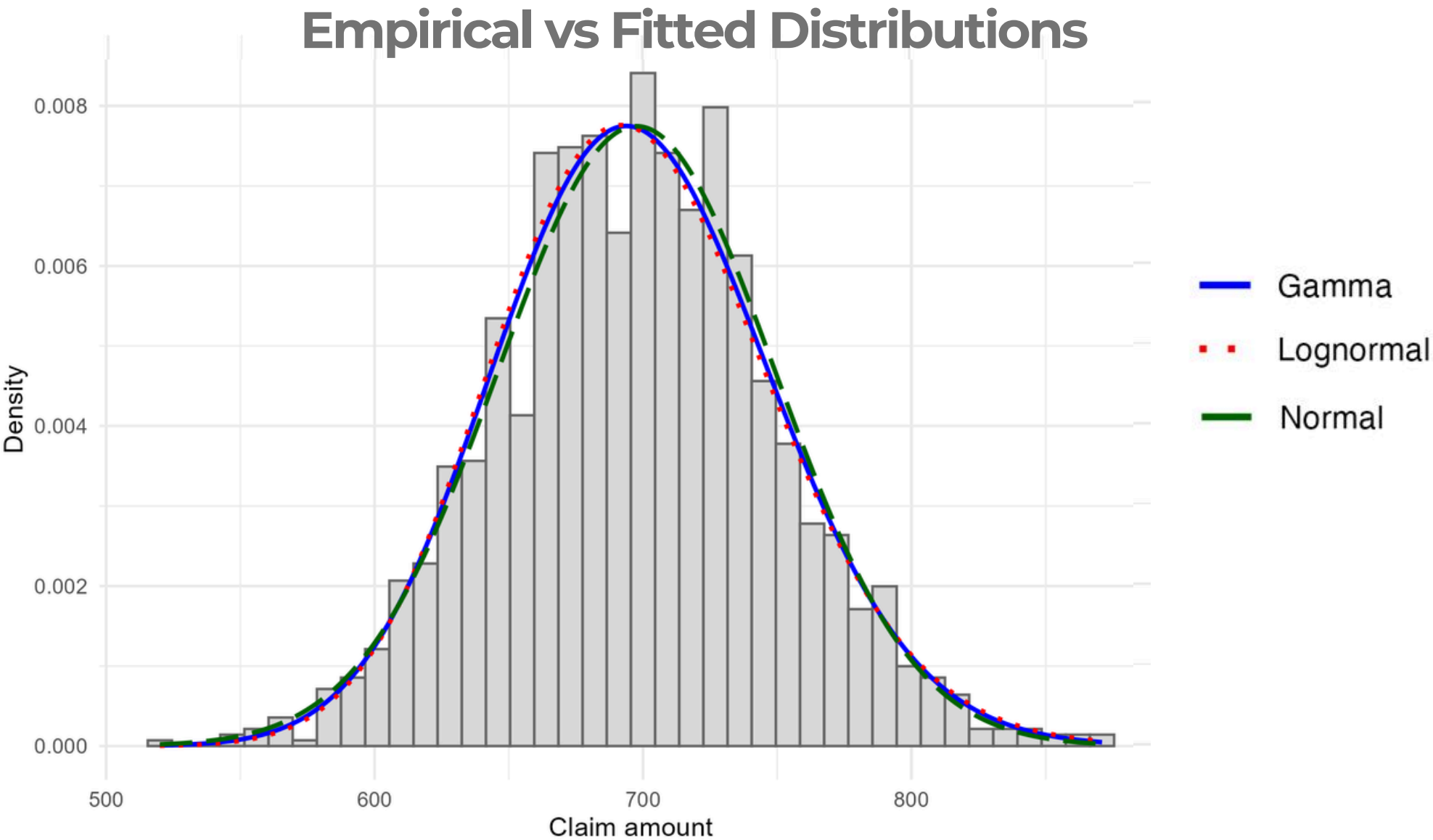


MODEL SELECTION & VALIDATION

Claim Severity

Distribution	AIC
Gamma	16717.17
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Exponential	23535.42
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Distribution	K-S p-value
Gamma	0.5816
Normal	0.4375
Lognormal	0.6491

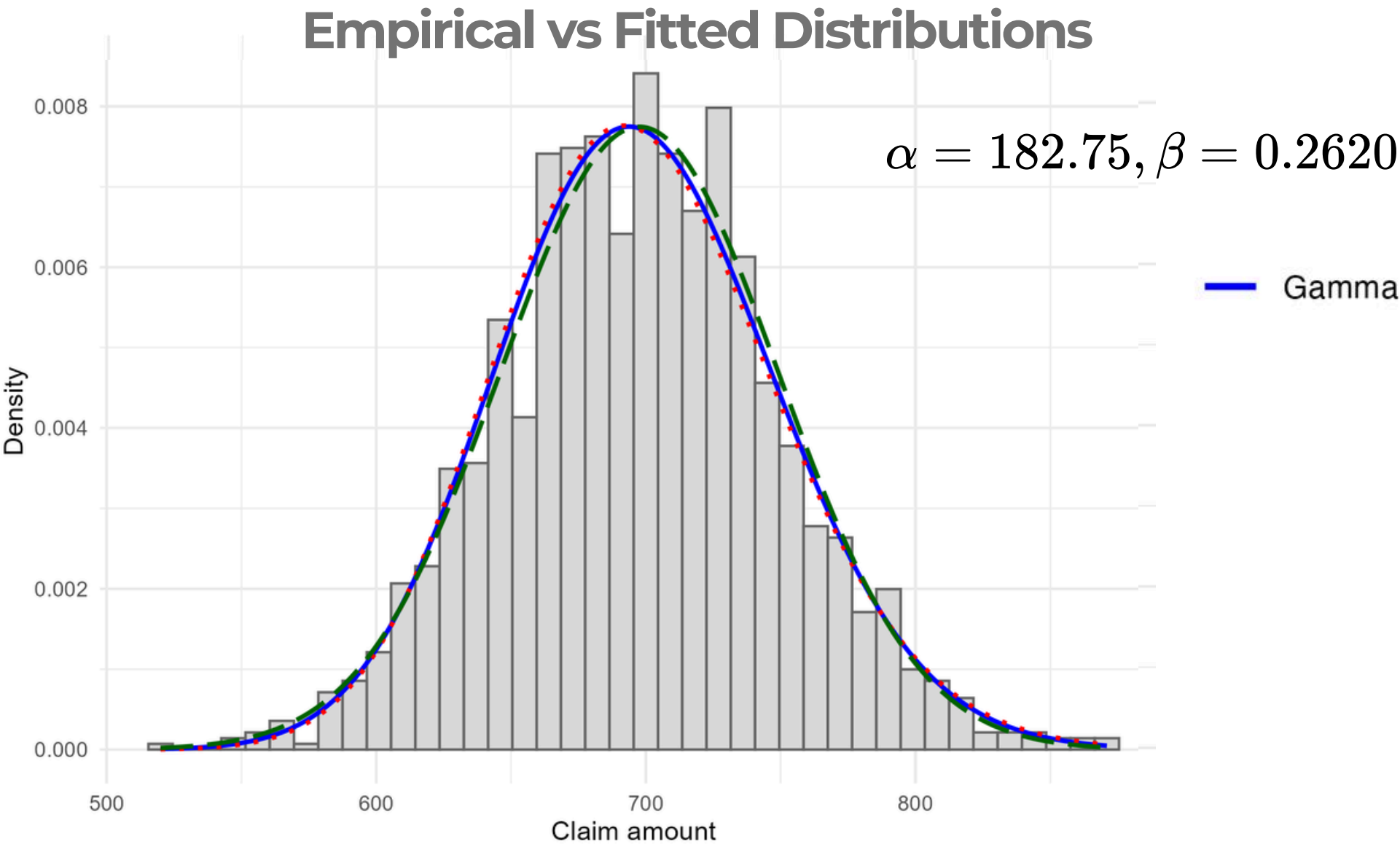


MODEL SELECTION & VALIDATION

Claim Severity

Distribution	AIC
Gamma	16717.17
Lognormal	16721.05
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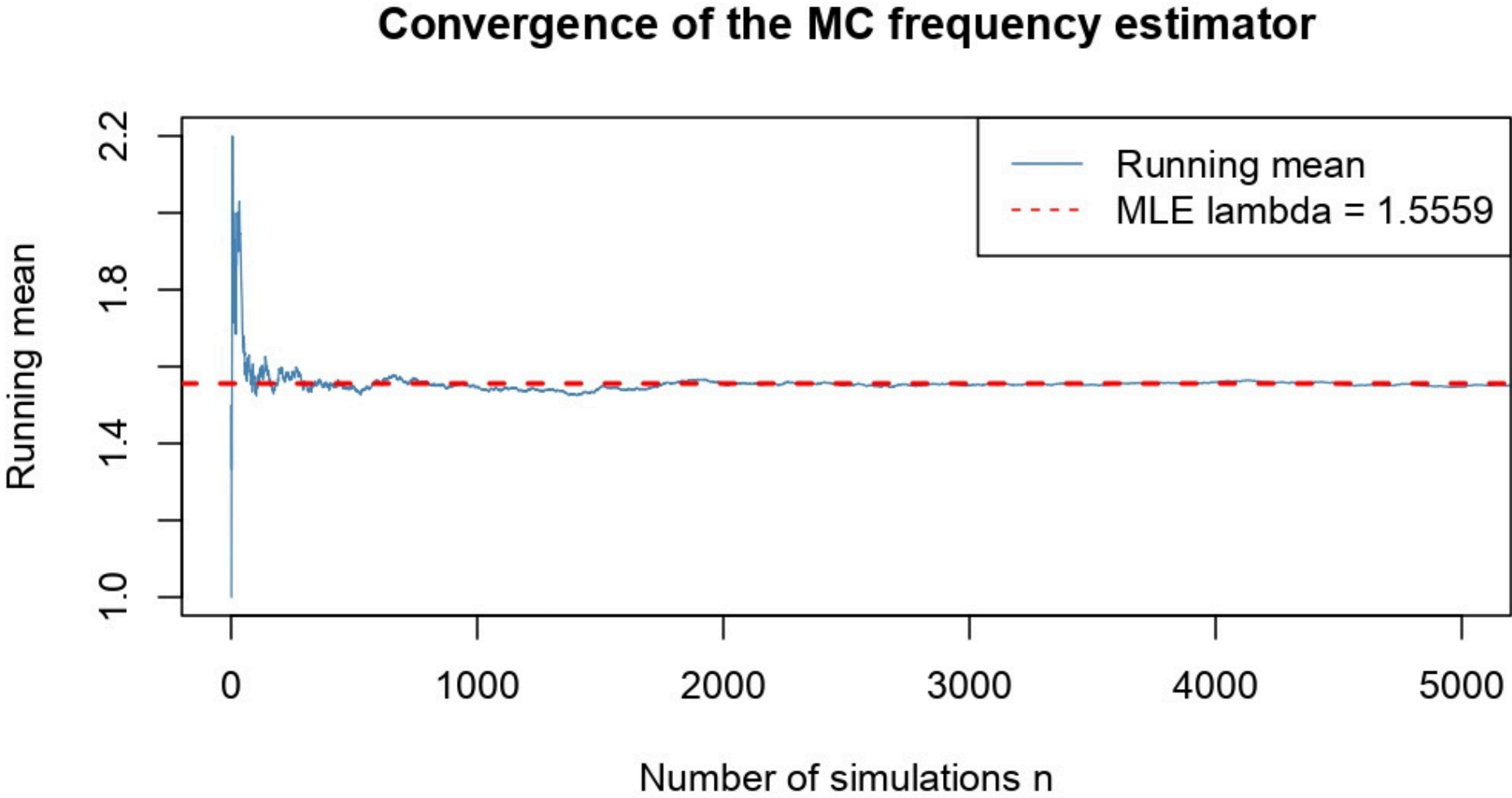
Distribution	K-S p-value
→ Gamma	0.5816
Normal	0.4375
Lognormal	0.6491



MODEL SELECTION & VALIDATION

Monte Carlo Estimation - Frequency

Monte Carlo Estimation	
Mean	1.5512
Variance	1.5524
Empirical Data	
Mean	1.5559
Variance	1.5737
Theoretical Model	
Mean	1.5559
Variance	1.5559

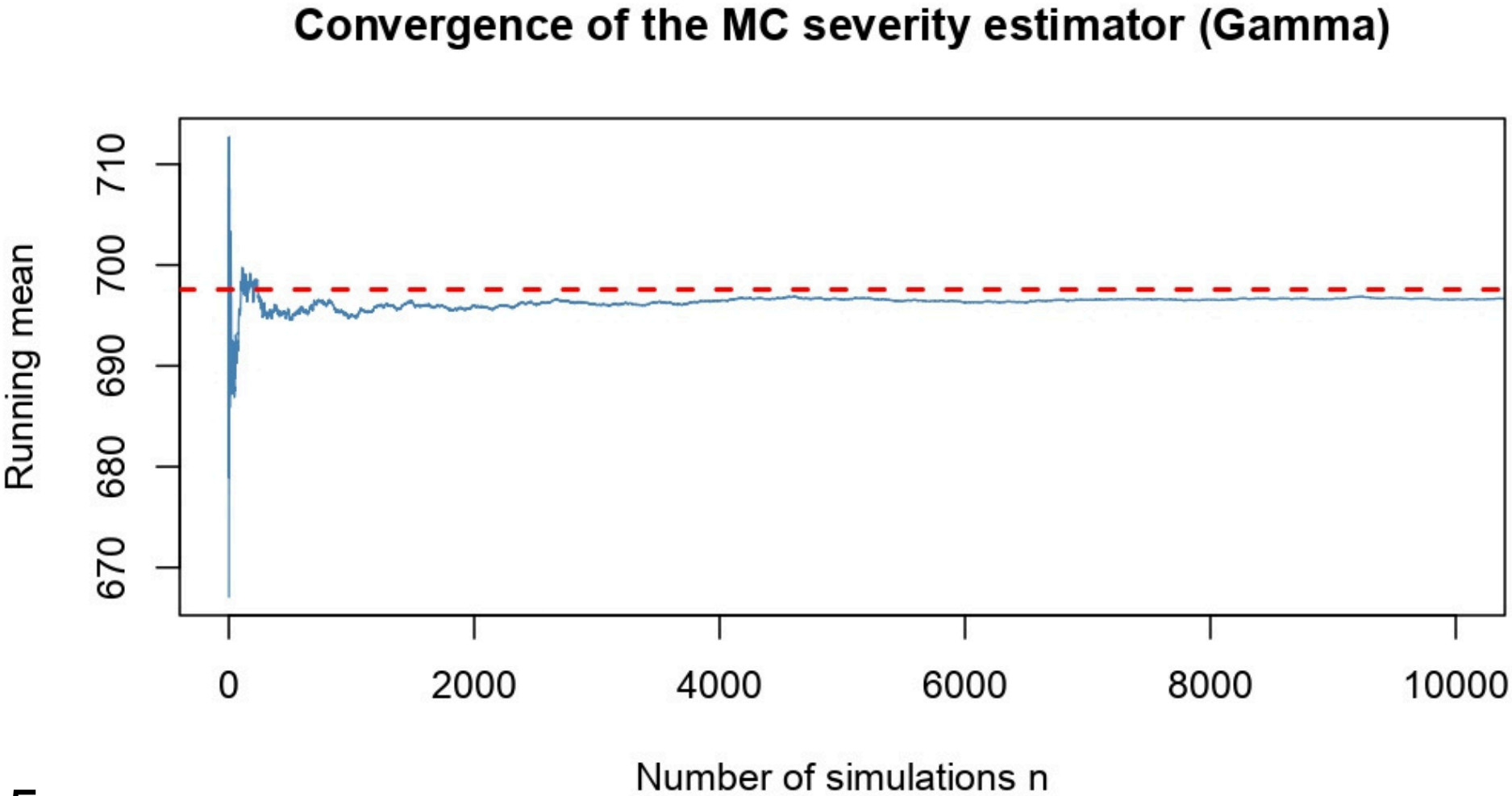


MSE of the Monte Carlo Estimator: 0.0000155

MODEL SELECTION & VALIDATION

Monte Carlo Estimation - Severity

Monte Carlo Estimation	
Mean	697.72
Variance	2,665.57
Empirical Data	
Mean	697.57
Variance	2,654.19
Theoretical Model	
Mean	697.57
Variance	2,662.65



MSE of the Monte Carlo Estimator: 0.02665

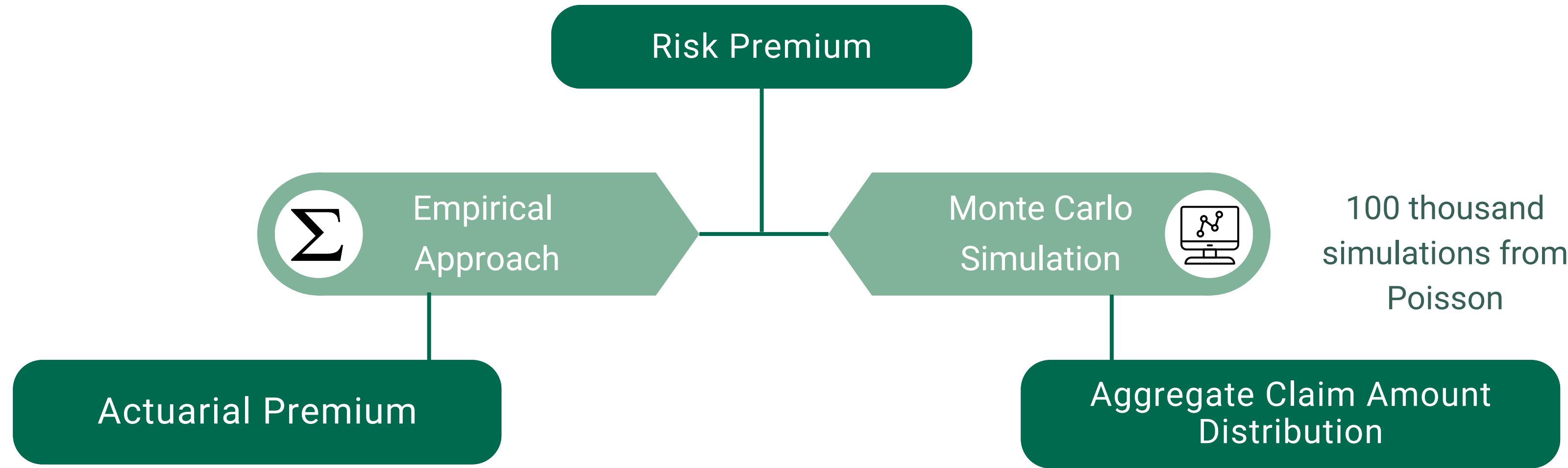
MODEL SELECTION & VALIDATION

Variance reduction techniques

Variable	Technique	Var (Crude)	Var (Reduced)	Var. Reduction	Correlation
Frequency	Antithetic	1.55×10^{-5}	2.10×10^{-6}	86.45%	-0.8641
Severity	Antithetic	2.67×10^{-2}	6.41×10^{-5}	99.76%	-0.9976
Severity	Control Variate	2.67×10^{-2}	3.18×10^{-5}	99.88%	+0.9994

- **Antithetic variates** : For a set of $n/2$ uniform draws, each draw is paired with its complement. Because these two are negatively correlated, their average has a much smaller variance and helps to balance out extreme values.
- **Control Variates**: For severity, the Normal distribution (fitted to the same data) is used as a control variate. It is almost perfectly correlated with the Gamma severity (due to the large shape parameter), so using it removes most of the variance.

APPLICATION



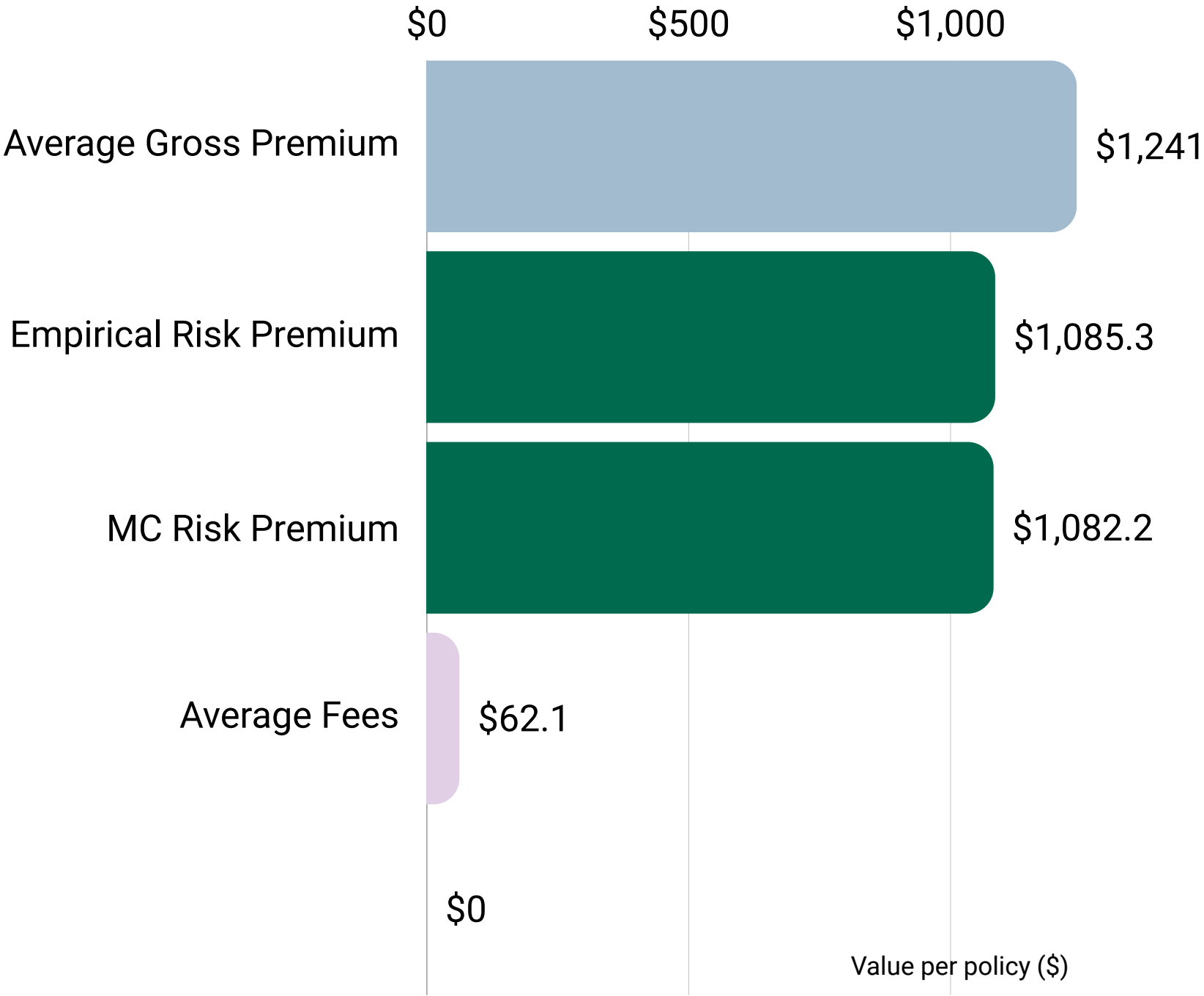
$$\pi = \frac{\sum \text{Claim Amount}}{\text{Number of Policies}}$$

$$E[S]$$

$$S = \sum_{j=1}^N Y_j \quad \begin{array}{l} Y_j \sim \text{Gamma}(\alpha, \beta) \\ N \sim \text{Poi}(\lambda) \\ N \perp (Y_j) \end{array}$$

APPLICATION

Risk Premium Comparison and Margin

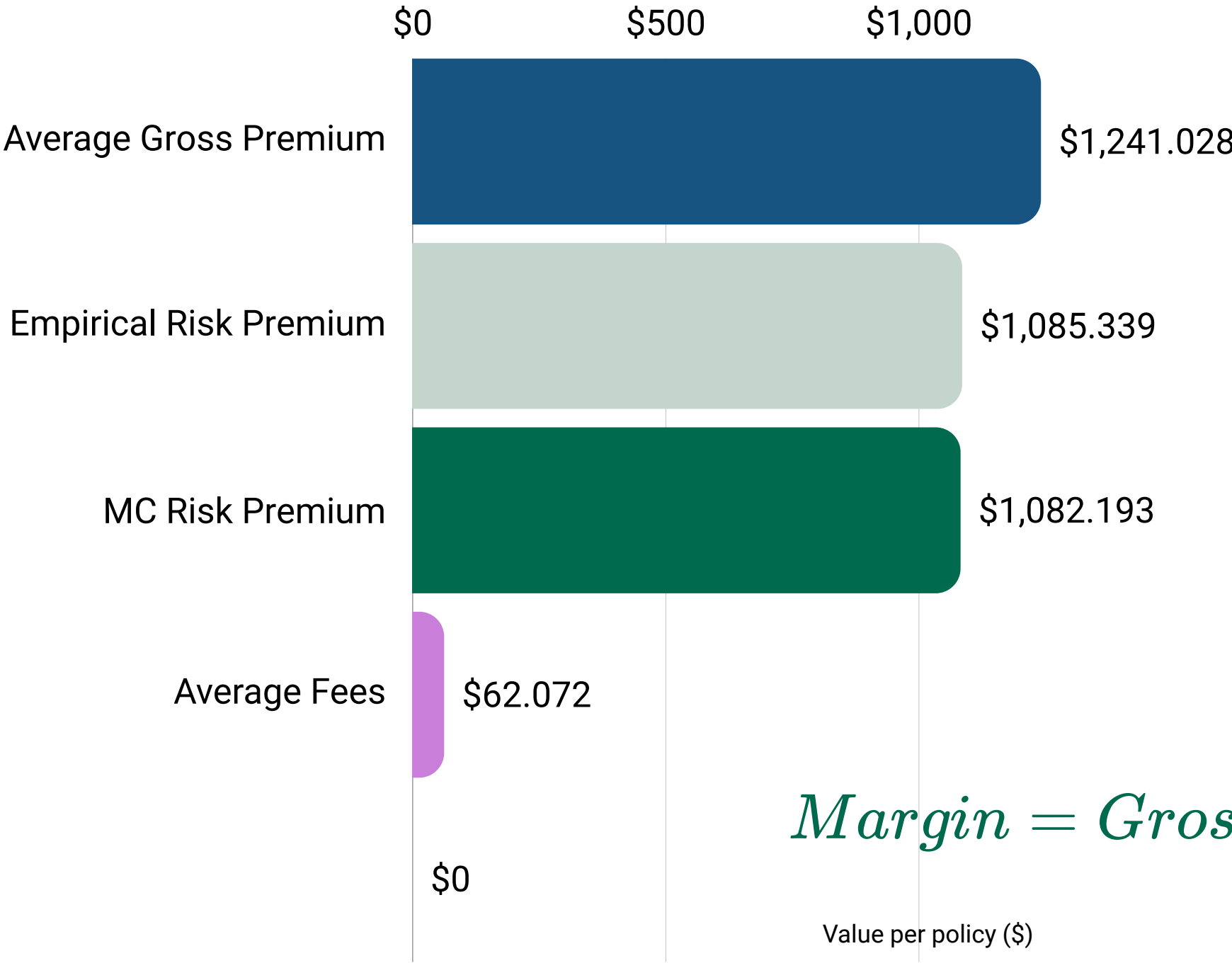


Comparison

The MC model closely matches the empirical risk premium, with a deviation below 0.3%, indicating a strong fit to the observed claims experience

APPLICATION

Risk Premium Comparison and Margin



Margin

Estimated margin of **\$96.76** per policy, or about 7.8% of the premium charged

$$\textit{Margin} = \textit{Gross Premium} - (\textit{Risk Premium} + \textit{Fees})$$

APPLICATION

Solvency Capital Requirements

Value at Risk

$$\text{VaR}_\alpha(S) = \inf\{x : F_S(x) \geq \alpha\}$$

It is the α -quantile for the loss distribution.

Method	Value
$VaR_{99.5\%}^{per\ policy}$	\$ 4,002.7
$VaR_{99.5\%}^{per\ portfolio}$	\$1,159,529
$E[S]$	\$1,087,431
SCR	\$72,098

APPLICATION

Solvency Capital Requirements

SCR

$$SCR = VaR_{99.5\%} - E[S]$$

Under Solvency II, it represents the amount of capital an insurer must hold to absorb severe losses and remain solvent with a 99.5% confidence level.

Method	Value
$VaR_{99.5\%}^{per\ policy}$	\$ 4,002.7
$VaR_{99.5\%}^{per\ portfolio}$	\$1,159,529
$E[S]$	\$1,087,431
SCR	\$72,098

REINSURANCE

Is it necessary?

- Purpose: protect against large losses, reduce volatility, support capital adequacy
- Indicator used: Combined Ratio + tail risk (99.5% VaR)

$$CR = \frac{\text{Total Claims} + \text{Administrative Costs}}{\text{Total Premium Paid}}$$

- $CR < 1 \rightarrow$ **profitable**
- $CR > 1 \rightarrow$ underwriting **loss**

current CR = 0.9246

In our case, reinsurance it's not strictly necessary *

MODEL EXPLANATION

IS THE PORTFOLIO **PROFITABLE?**

MODEL EXPLANATION

IS THE PORTFOLIO **PROFITABLE**?

HOW MUCH CAN WE **REDUCE** THE TARIFF?

MODEL EXPLANATION

IS THE PORTFOLIO **PROFITABLE**?

HOW MUCH CAN WE **REDUCE** THE TARIFF?

SHOULD THE REDUCTION BE **UNIFORM**?

TARIFF SENSITIVITY

- Overall portfolio

Tariff Reduction	0%	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
Combined Ratio	0.92	0.93	0.94	0.95	0.96	0.97	0.98	0.99	1.01	1.02	1.03



TARIFF SENSITIVITY

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Combined Ratio	0.92	0.93	0.94	0.95	0.96	0.97	0.98	0.99	1.01	1.02	1.03

- Segmented by vehicle type

Tariff Reduction	0%	1%	2%	3%	4%	5%	6%	7%	8%	9%	10%
SUV	0.96	0.97	0.98	0.99	1.00	1.01	1.02	1.03	1.04	1.06	1.07
Non-SUV	0.91	0.92	0.93	0.94	0.95	0.96	0.97	0.98	0.99	1.00	1.01

TARIFF SENSITIVITY

- Overall portfolio

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Non-SUV	0.91	0.92	0.93	0.94	0.95	0.96	0.97	0.98	0.99	1.00	1.01

Differentiated tariffs better reflect the true risk profile

KEY TAKEAWAYS

- **Pricing adequacy and profitability**

Portfolio is profitable, current margin of \$97 per policy

- **Tariff flexibility**

Feasible to decrease uniformly up to 5%, capped at 2% for SUVs

- **Reinsurance position**

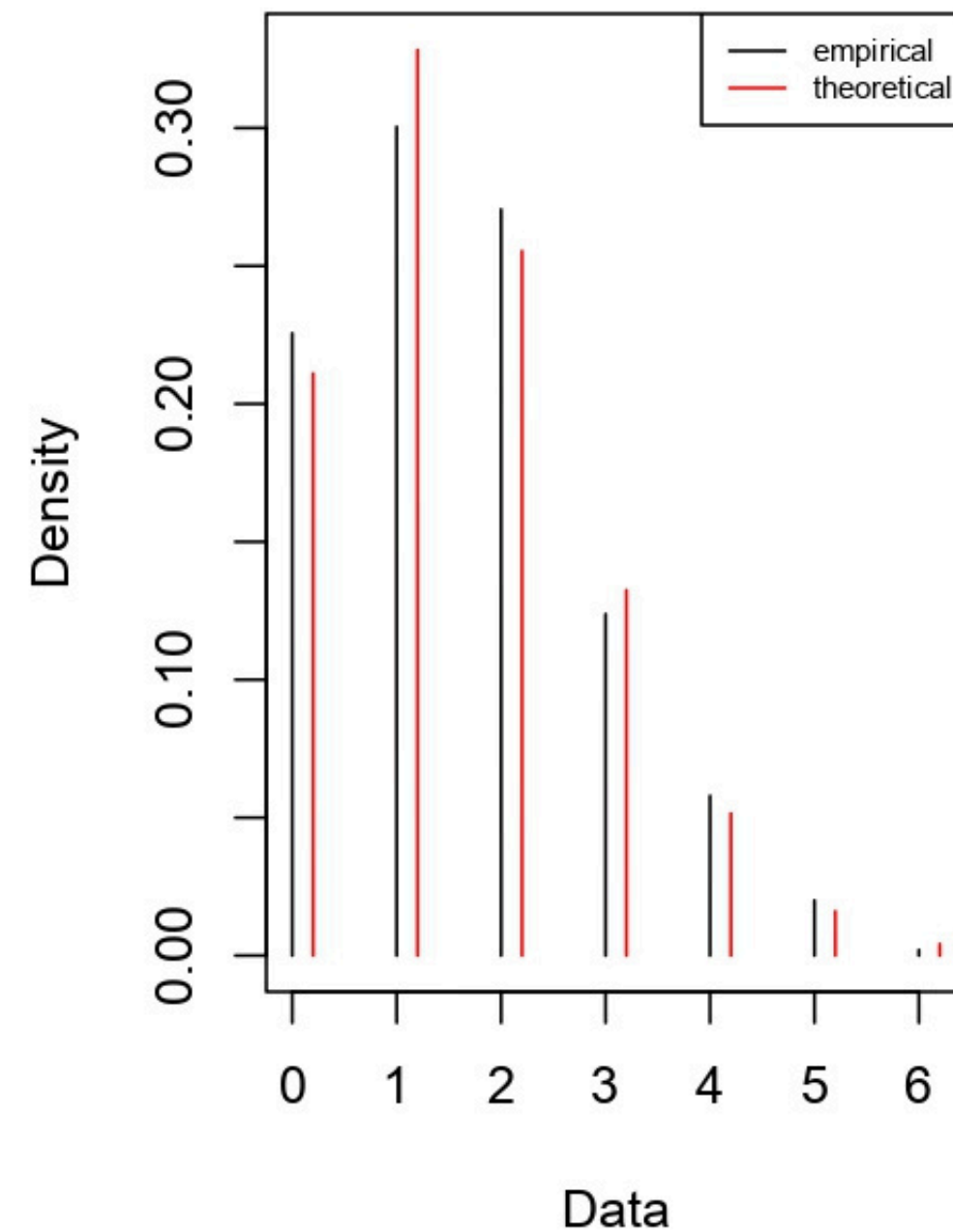
Optional, Excess of Loss contract as a prudent safeguard

THANK YOU!
ANY QUESTIONS?

MODEL SELECTION & VALIDATION

Claim Frequency

Emp. and theo. distr.



Emp. and theo. CDFs

